

# Euclid's Elements — Book I

## Proposition 34

Formal Reconstruction — Technical Documentation

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*In parallelogrammic areas the opposite sides and angles are equal to one another, and the diameter bisects the areas.*

## 1 Introduction

Euclid's Proposition I.34 collects the fundamental metric and angular properties of a parallelogram. For a parallelogram  $ABCD$ , it asserts that opposite sides are equal, opposite angles are equal, and a diagonal divides the parallelogram into two equal parts.

The present reconstruction separates these three assertions according to the language actually available in the CGJteamLab Hilbert development. The first two are represented directly by congruence of segments and angles. The final claim is represented by congruence of the two triangles cut off by the diagonal. No numerical area function is introduced.

This Proposition is therefore a useful example of a recurring distinction in the project: the classical theorem and the formal theorem have the same synthetic content, but the Lean proof uses a different proof architecture from Euclid's original argument.

## 2 The Classical Configuration

Let  $ABCD$  be a parallelogram, with

$$AB \parallel CD, \quad BC \parallel DA.$$

Draw the diagonal  $AC$ . The diagonal separates the parallelogram into the triangles  $ABC$  and  $CDA$ .

The required conclusions are

$$\begin{aligned} AB &\cong CD, & BC &\cong DA, \\ \angle DAB &\cong \angle BCD, & \angle ABC &\cong \angle CDA, \end{aligned}$$

and, in the language of the present formalization,

$$\triangle ABC \cong \triangle CDA.$$

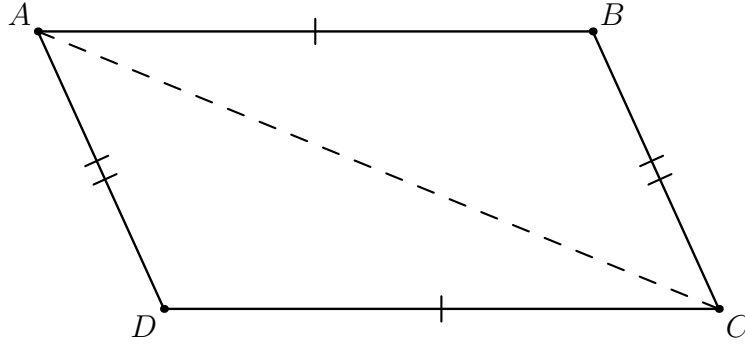


Figure 1: The configuration for Proposition I.34. The quadrilateral  $ABCD$  is a parallelogram and  $AC$  is the diagonal used in the reconstruction.

### 3 Classical Proof

Euclid’s proof uses the diagonal and the theory of parallel lines established immediately before Proposition I.34. Since  $AB \parallel CD$ , Proposition I.29 gives one pair of alternate angles. Since  $BC \parallel DA$ , a second application of Proposition I.29 gives another pair of alternate angles. The common diagonal then allows Proposition I.26 to identify the two triangles. The corresponding opposite sides and angles are therefore equal. The final statement that the diagonal bisects the parallelogram follows because the two triangles are congruent.

A compact classical dependency graph is

$$\text{I.29} + \text{I.29} \longrightarrow \text{I.26} \longrightarrow \begin{array}{l} \text{opposite sides equal,} \\ \text{opposite angles equal,} \\ \text{diagonal gives equal triangles.} \end{array}$$

The Beeson–Narboux–Wiedijk proof file makes explicit several configuration facts hidden by the classical diagram: non-collinearity of the relevant triangles, opposite-side data for the transversal configuration, and endpoint order needed to apply the formal versions of I.29 and I.26. These details are part of the classical reconstruction problem, even though they are not visible in Euclid’s short paper proof.

### 4 The Main Formalization Issue: Reusing Parallelogram Theory

The current CGJteamLab proof does not replay the classical I.29–I.26 route. By the time Proposition I.34 is reached, the Hilbert interface already contains a reusable Euclidean theory of parallelograms. In particular, it contains a theorem stating that the opposite sides of a Euclidean parallelogram are congruent.

This changes the proof DAG substantially. Once

$$AB \cong CD \quad \text{and} \quad BC \cong DA$$

are available, the diagonal  $AC$  supplies the third side congruence

$$AC \cong CA.$$

The triangles  $ABC$  and  $CDA$  are therefore congruent by the already proved Hilbert SSS theorem. The required opposite-angle congruence is extracted from that triangle-congruence result.

Thus the formal proof uses the same diagonal as Euclid, but it enters the triangle argument through SSS rather than through two alternate-angle relations and ASA.

## 5 Proof Architecture of the Reconstruction

The reusable proof architecture is

$$\begin{array}{c}
 AB \parallel CD, \quad BC \parallel DA \\
 \Downarrow \\
 \text{IsParallelogram } ABCD \\
 \Downarrow \\
 AB \cong CD, \quad BC \cong DA \\
 AC \cong CA \\
 \Downarrow \text{SSS} \\
 \triangle ABC \cong \triangle CDA \\
 \Downarrow \\
 \angle ABC \cong \angle CDA.
 \end{array}$$

The second pair of opposite angles is obtained by applying the same reusable angle theorem to the cyclically relabelled parallelogram  $BCDA$  and then using symmetry of angle congruence.

This produces two interface-level helpers:

```

ParallelogramOppositeAngleCongruent
ParallelogramOppositeAnglesCongruent

```

The first proves one pair of opposite angles. The second packages both pairs. They belong to `HilbertInterface` because they are general properties of every Euclidean parallelogram, not facts specific to Proposition I.34.

## 6 Lean Formalization

The main Proposition theorem is deliberately thin:

```

theorem euclid_proposition_34
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D : Geo.Point)
  (hParallelogram : IsParallelogram Geo A B C D) :
  And
    (OppositeSidesCongruent Geo A B C D)
    (OppositeAnglesCongruent Geo A B C D)

```

Its two branches are discharged by the established interface theorems

```

ParallelogramOppositeSidesCongruent
ParallelogramOppositeAnglesCongruent

```

The third part of Euclid’s statement is represented separately:

```
theorem euclid_proposition_34_diagonal
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D : Geo.Point)
  (hParallelogram : IsParallelogram Geo A B C D) :
  TriangleCongruenceResult Geo A B C C D A := by
```

The proof first derives non-collinearity of  $A, B, C$  from the disjointness of the parallel lines  $AB$  and  $CD$ . It then obtains the two pairs of opposite side congruences from the parallelogram theorem, uses reflexivity of congruence for the diagonal, and applies SSS:

```
have hSides :
  OppositeSidesCongruent Geo A B C D :=
  ParallelogramOppositeSidesCongruent
  Geo A B C D hParallelogram

have hAC_CA :
  Geo.Congruent A C C A :=
  CongruentSwapSecond
  Geo A C A C
  (hilbert_congruent_reflexive Geo A C)

have hSSS :=
  HilbertSSS
  Geo
  A B C
  C D A
  hABC
  hSides.1
  hSides.2
  hAC_CA

exact hSSS.2
```

No area object occurs in this theorem. The formal counterpart of Euclid’s “diameter bisects the areas” is the congruence of the two triangles cut off by the diagonal.

## 7 Relationship with Earlier Propositions

Classically, Proposition I.34 depends directly on I.29 and I.26. In the present Lean development these earlier Euclidean consequences have already been absorbed into the reusable parallelogram theory of `HilbertInterface`. Consequently the final Proposition file does not call `euclid_proposition_29` or `euclid_proposition_26` directly.

This is an intentional architectural compression. The classical dependency has not disappeared mathematically; it has been moved into general Hilbert-level theorems that are proved once and then reused.

The relation with I.33 is especially close. Proposition I.33 recognized a parallelogram from one pair of opposite sides that were both parallel and congruent, together with orientation data.

Proposition I.34 starts after the quadrilateral has already been recognized as a parallelogram and extracts its standard Euclidean consequences.

## 8 Hilbert and Book Zero Components Used

The formal Proposition uses the following established components:

- `IsParallelogram`, defined by two pairs of opposite parallel sides;
- `ParallelogramOppositeSidesCongruent`, the Euclidean theorem giving the two opposite-side congruences;
- `HilbertSSS`, the derived SSS congruence theorem;
- incidence uniqueness and the extensional representation of parallel lines, used to prove that  $A, B, C$  are non-collinear;
- reflexivity and endpoint symmetry of segment congruence;
- symmetry of angle congruence, used when packaging the second pair of opposite angles.

No new Book Zero theorem is required for Proposition I.34.

## 9 Formalization Notes

### 9.1 Axiomatic status

Proposition I.34 is Euclidean in the present formulation. The hypothesis is

```
[HilbertEuclideanPlane Geo]
```

The Euclidean strength is consumed in the general theorem asserting congruence of opposite sides of a parallelogram. This is not a neutral fact for a parallelogram defined only by parallel opposite sides: in hyperbolic geometry such a quadrilateral need not have congruent opposite sides.

After those side congruences are available, the local SSS step itself belongs to the congruence theory and does not require an additional use of the parallel axiom.

### 9.2 Hidden configuration data

The concise classical diagram suppresses a non-degeneracy issue. Lean must show explicitly that  $A, B, C$  are not collinear before applying SSS. The proof uses the fact that  $AB$  and  $CD$  are parallel and hence their extensional point-lines are disjoint. If  $C$  lay on  $AB$ , then  $C$  would belong to both parallel lines, a contradiction.

No new `SameSide`, `OppositeSide`, or `Between` witness is needed in the final Proposition I.34 proof. Those notions occur in the lower-level derivation of the reusable Euclidean parallelogram theory, but the API-level proof has already hidden them behind the established theorem.

### 9.3 Reusable helpers introduced

Two reusable parallelogram helpers were added while proving I.34:

```
ParallelogramOppositeAngleCongruent
ParallelogramOppositeAnglesCongruent
```

The first derives one pair of opposite angles from SSS on the two triangles cut by a diagonal. The second obtains both pairs by cyclic relabelling. These are not local Proposition helpers: they belong to `HilbertInterface` and are available to later propositions.

The packaging predicate

```
OppositeAnglesCongruent
```

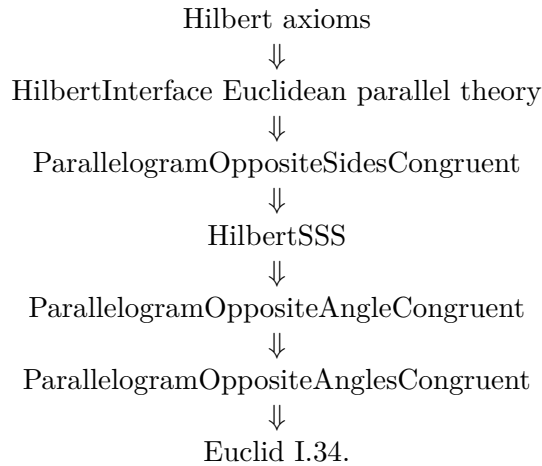
records the two required angle congruences.

## 10 Dependency Summary

The classical dependency is

$$\text{I.29} + \text{I.29} \longrightarrow \text{I.26} \longrightarrow \text{I.34}.$$

The actual Lean dependency is



For the diagonal theorem the last part is even shorter:

$$\text{opposite sides congruent} + AC \cong CA \longrightarrow \text{SSS} \longrightarrow \triangle ABC \cong \triangle CDA.$$

|                         |     |
|-------------------------|-----|
| New axiom:              | no  |
| New reusable helper:    | yes |
| Book Zero helper added: | no  |
| Build clean:            | yes |
| sorry:                  | no  |

## 11 Conclusion

Proposition I.34 is short at the Proposition-file level because the general Euclidean theory of parallelograms has already been established in the Hilbert interface. This is a deliberate gain from the layered architecture of the project.

Euclid obtains the theorem from parallel-angle relations and ASA. The present Lean reconstruction instead reuses the already derived congruence of opposite sides and applies SSS across the diagonal. The resulting proof is not a mechanical transcription of Euclid, but it preserves the same synthetic geometry while exposing where the Euclidean content actually resides.

The third classical assertion is represented without introducing area: the diagonal  $AC$  is proved to cut the parallelogram into the congruent triangles  $ABC$  and  $CDA$ .